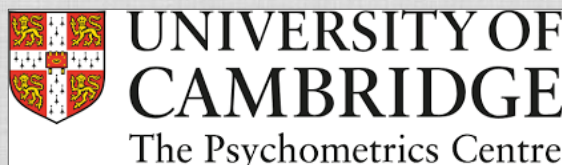


Where is the data come from?

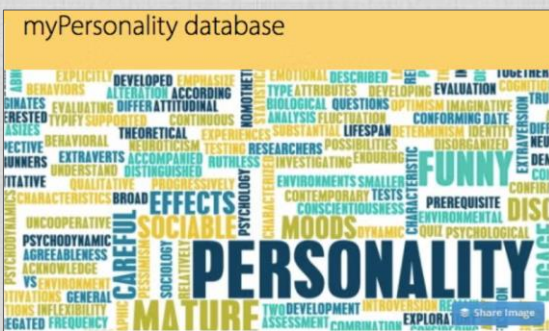
And who collected it?

Authors

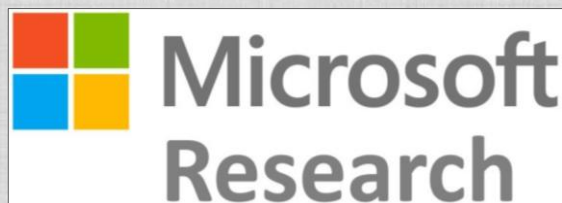
Michal
Kosinski



David
Stillwell



Thore
Graepel



Data

58,466



Features

Users' Facebook Likes

55,814 Likes

	art	cnn.com	(...)	BMW
User 1	1	1	...	0
User 2	0	1	...	1
User 3	1	0	...	0
(...)	...	...	...	...
User n	1	1	...	0

58,466 Users

User – Like Matrix
(10M User-Like pairs)

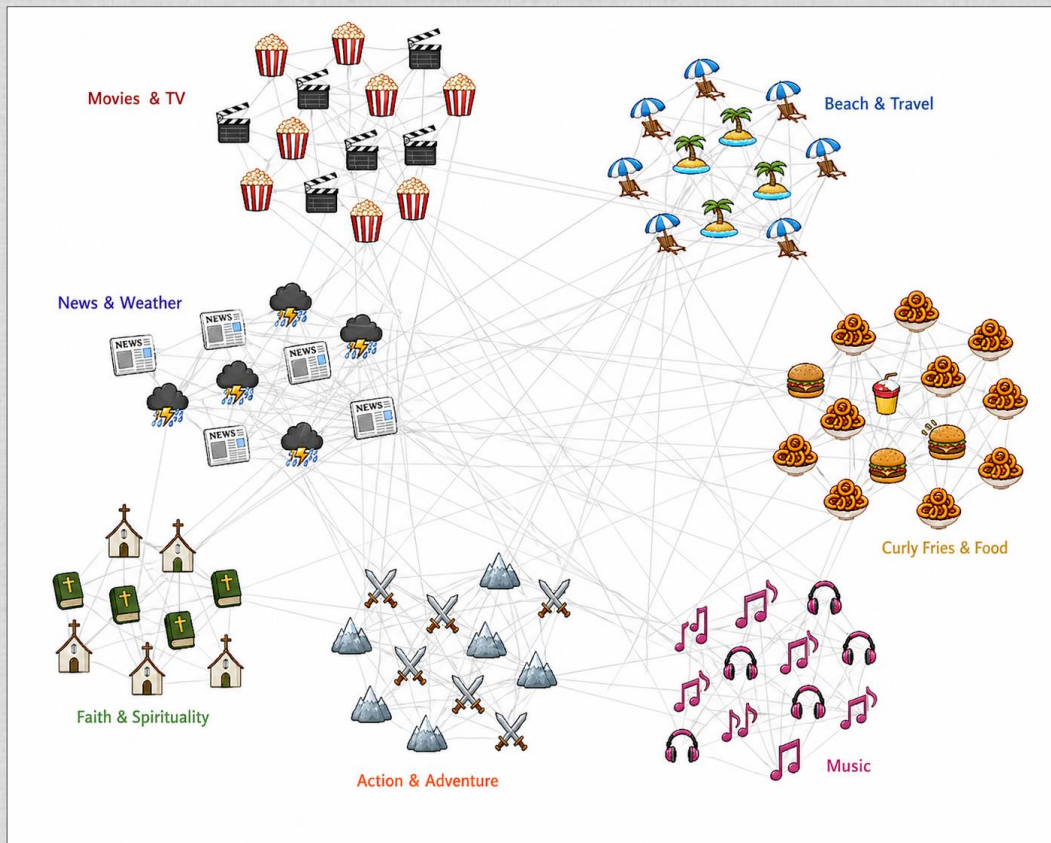
Labels

- FFM personality Score
- Intelligence
- SWL Scale
- Age
- Gender
- Relationship
- Political views
- Religion
- Facebook
- Consumption(alc./drug/cig.)
- Parents Staying Together until 21
- Ethnic Origin
- Sexual Orientation

Not Representative: Quality Issues in Collecting/Sampling

❑ “Likes” collected are NOT random.

NOT -> people like curly fries



BUT -> only certain people were given curly fries to like

❑ Being “Random” is NOT absolute.

$$P(Y \mid \text{Collecting}) = P(Y)$$

Select “Volunteered” as filter

Collecting	Y Response	SVD1	SVD2	SVD3	...	SVDk
☒ Volunteered	1	2.31	-1.12	0.87	...	-0.45
☐ Others	1	1.95	-0.98	1.02	...	0.33
☒ Volunteered	0	2.44	-1.30	0.76	...	-0.21
☒ Volunteered	1	2.10	-1.05	0.91	...	0.11
☒ Volunteered	0	1.88	-0.89	1.15	...	-0.37
☐ Others	0	0.12	0.45	-0.33	...	0.08
☐ Others	1	-0.08	0.39	-0.41	...	-0.02
☐ Others	0	0.05	0.51	-0.29	...	0.03
☐ Others	1	-0.11	0.47	-0.36	...	-0.01
☐ Others	0	0.09	0.42	-0.31	...	0.04

Included (Volunteered only)

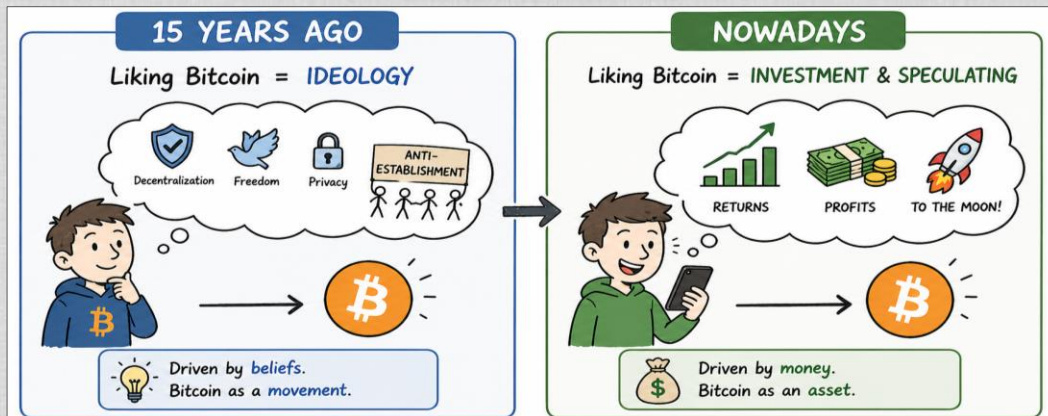
Excluded (Others filtered out)

Transfer learning is also useful to handle multi-tasks with the same dataset.

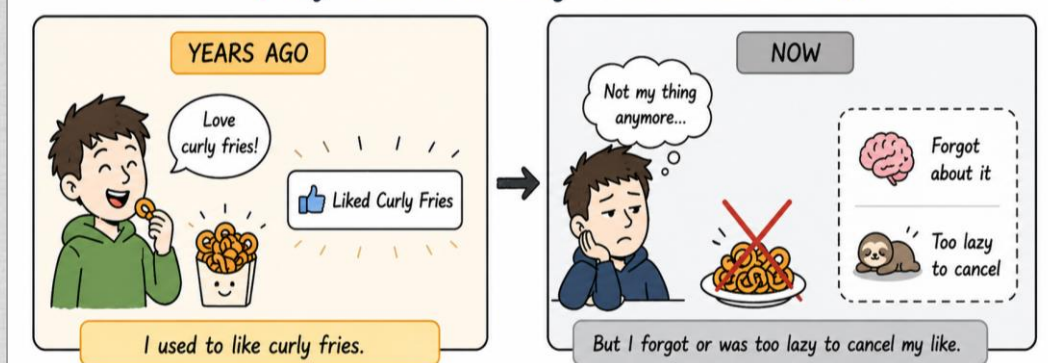
Not Representative: Quality Issues in Feature Proxies

□ Temporal instability of “Likes”.

Shifting in proxy meaning



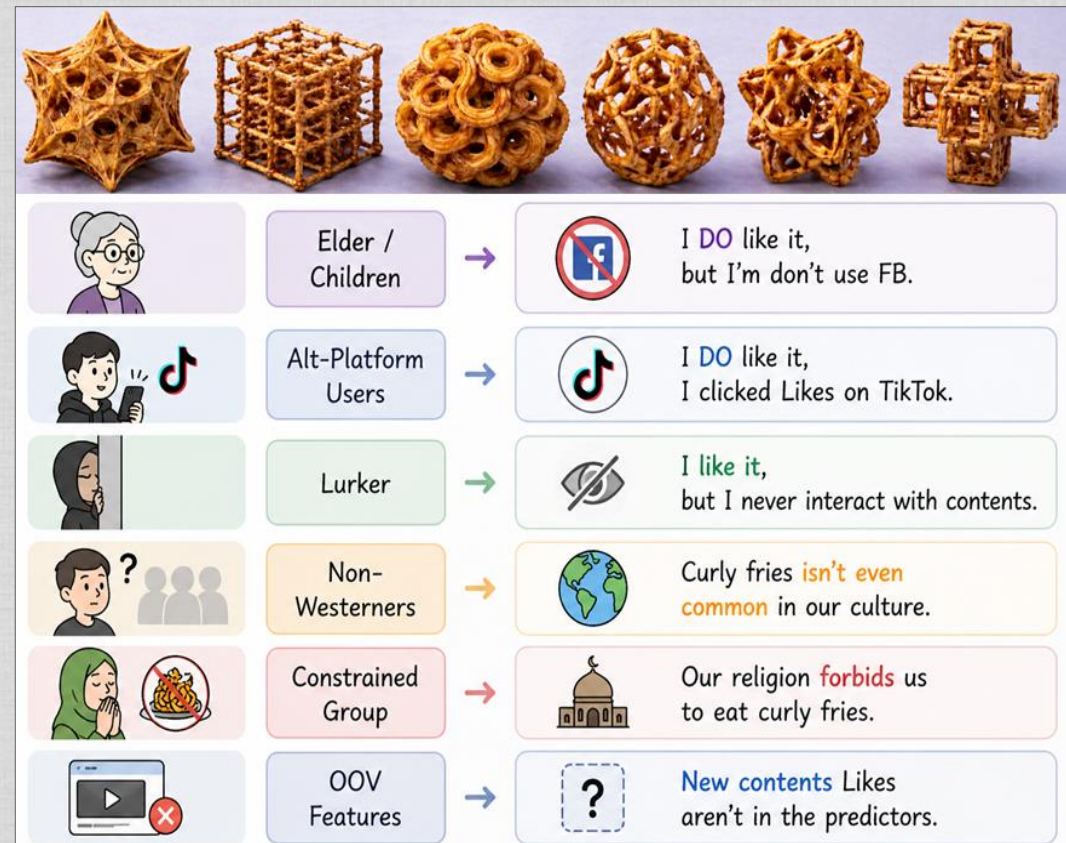
My “Like” for Curly Fries is Outdated



Shifting in user preference

□ Silenced groups.

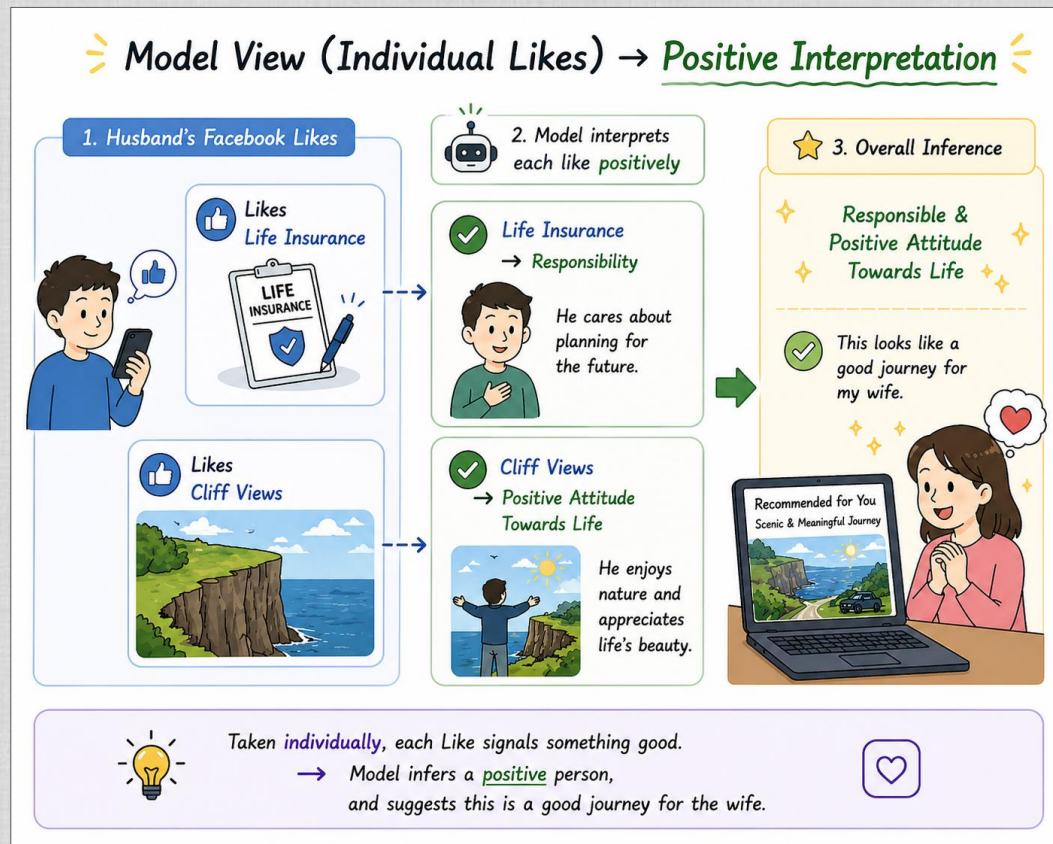
Absence of Like $\neq$ Absence of Preference



Not Representative: Quality Issues in Feature Embedding

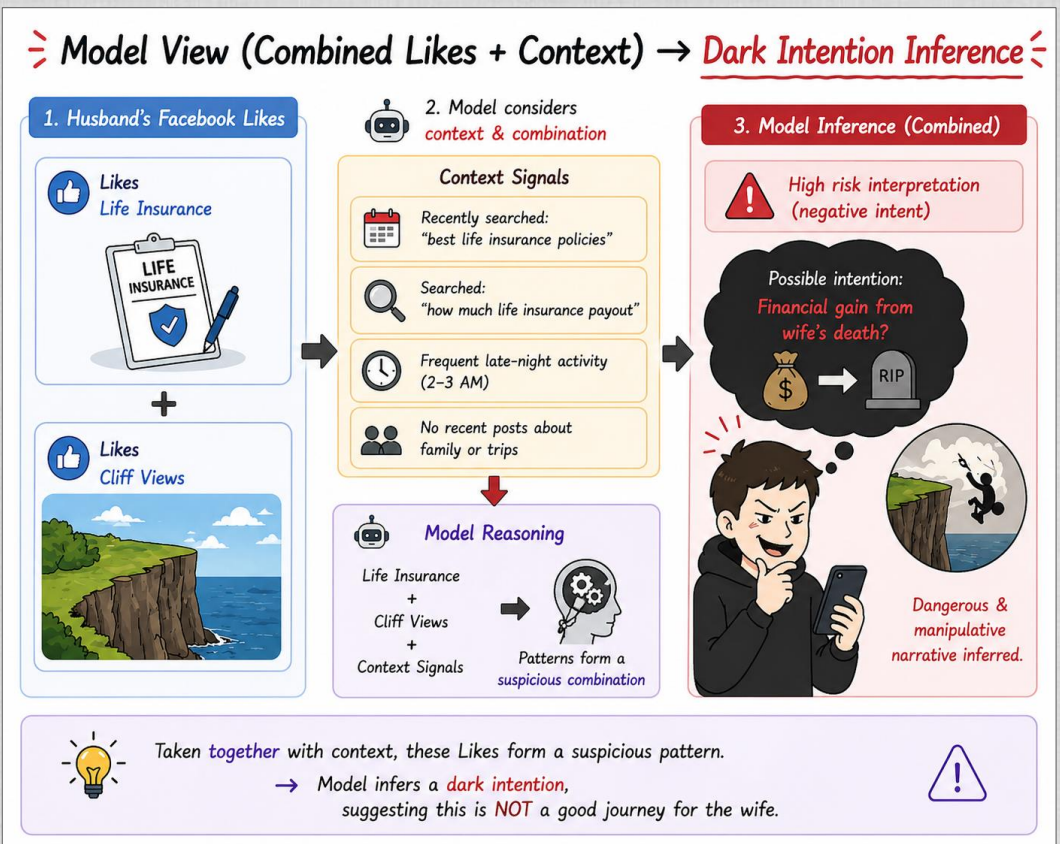
❑ Static embedding (SVD)

Static Embedding **CANNOT** handle contexts



❑ Potential contextual embedding

"Life Insurance" + "Cliff" = Worst case ever



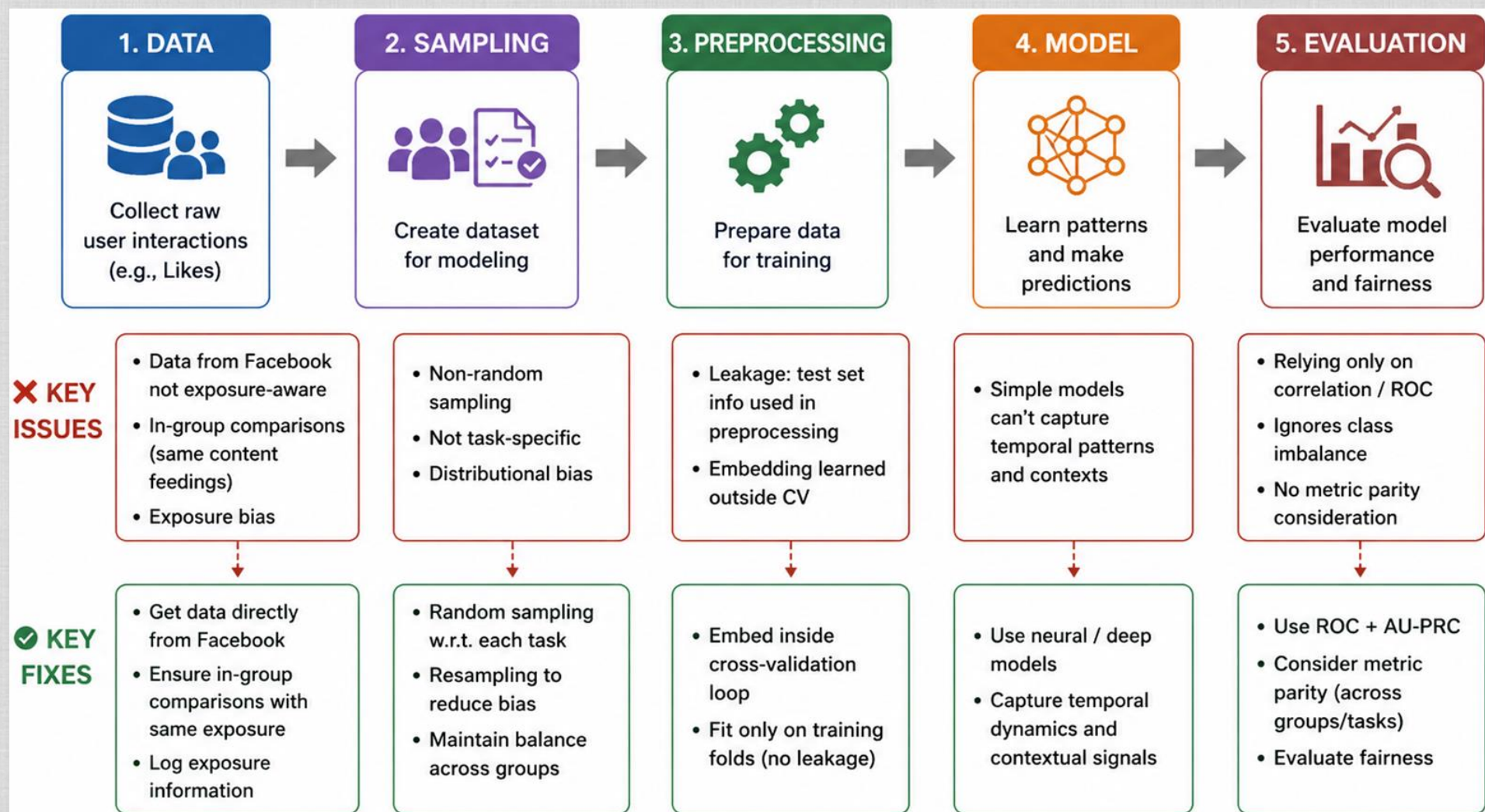
Not Representative: Quality Issues in **Label Measurement**

❑ Subjective, measurement shifting, and minor issues (inaccuracy, diversity)

Problem Type	Why It Fails	Example Label	Collection Clue
Self-reporting Bias	Users may hide, fake, or not update	Sexual Orientation	"Interested in" field — Facebook profile
		Relationship Status	"Single" / "In relationship"
		Alcohol Consumption	Self-reported (users)
		Drug Use	Self-reported (users)
Latent & Approximate Measures	True traits are unobservable, proxy label were used instead	FFM Personality Score	Questionnaire
		Intelligence	Raven's Standard Progressive Matrices (SPM)
		SWL Scale	Potential subjective user reaction on "How do you feel today?"
		Ethnic Origin	Visual inspection of profile pictures (manual)
Inaccurate Defination	Heterosexual was miss classified	Heterosexual	miss bi-sexual when defining (those interested in users of the opposite gender were labeled as heterosexual)
Missing Diversity	Real-world variation is excluded	Gender Binary only	Predefined from profile "Interested in" field
		Ethnic Origin	73% Caucasian, 14% African American, while 13% other not represented

Technical issues that are preventing model from deployment.

GOAL: Unbiased data -> Reliable models -> Fair and trustworthy deployment



Practical issues that are preventing model from deployment.

The model fails to generalise - across meaning, population, time, and scale.

1 From Association → Causation



- Example: curly fries meme
- Model learns correlation (what happens together), not cause (why it happens)



Correlation is mistaken as causation.

2 Out-of-Sample Population



- Example: non-US, non-Westerners, minority ethical / religious groups
- Model is trained on a limited population



Model does not represent unseen populations.

3 Future Data (Temporal Shift)



- Example: new likes on new contents
- Old model cannot capture new patterns and trends



The world changes, but the model doesn't.

4 Small Groups (Data Scarcity)



- Example: groups with few observations
- Model memorizes noise ($df < 0$) → leads to bias



Less data → more bias.

Moral issues that are preventing model from deployment.

These system don't just predict, they **shape**, **amplify**, and **harm**.

1 Reinforces & Polarizes User Behaviors

Example: Cambridge Analytica



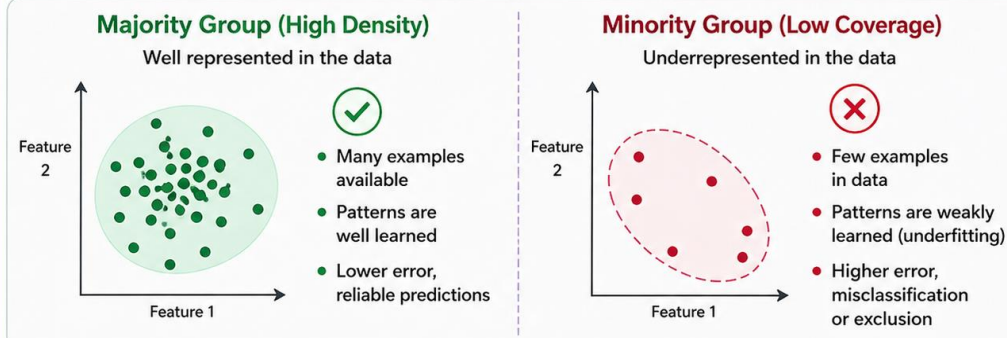
- Algorithms decide what users see, repeatedly.
- This nudges beliefs, emotions, and even election outcomes.
- Creates echo chambers and deepens social divisions.



From predicting behavior →
to shaping and amplifying it.

2 Models Learn Density, Not Coverage

They fit what is frequent, not what is fully represented.



What the Model is Actually Doing

- Optimizes performance on frequent patterns (dense regions).
- Ignores or poorly fits rare or underrepresented groups.
- Learns **data density**, not true population coverage.



Why This Matters

- Underrepresented groups receive worse predictions.
- Leads to systematic bias and unfair outcomes.
- Can cause real-world harm.

Real-world risks (examples):



Overestimating crime risk



Unfair loan rejections



Misclassification in sensitive decisions



If a group is not well represented in the data, the model will not learn them well — leading to higher error and potential harm.

What must be fixed first?

In Believing: technology makes the world a better place.

To trust the system and deploy, we **need**



Data-Wise

- ✗ Likes collected regardless exposures;
- ✓ Likes collected from the same group where contents are fed;
- ✗ Universal sampling for multi-tasks;
- ✓ Task-specific sampling to secure randomness;

Preprocessing-Wise

- ✗ Global SVD before Cross Validation;
- ✓ Fit SVD inside CV, using trainset only;
- ❓ No resampling to solve distributional bias;
- ✓ Use resampling when needed to favor coverage than density;

Model-Wise

- ✗ Simple linear/logit models;
- ✓ Use models that can account for temporal patterns and contexts;
- ✓ Use transfer learning to leverage on pre-trained patterns for multi tasks;

Evaluation-Wise

- ❓ Used rigid ROC-AUC for classification;
- ✓ Sometimes, need AU-PRC to favor recall;
- ❓ No metrics parity consideration;
- ✓ Sometimes, need metrics parity for vulnerable groups;

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